

# Rapid Depolymerization of Polyester Fibers: Dimethyl Carbonate-Aided Methanolysis Combined with the Ball-Milling Approach

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Cite This: *ACS Sustainable Chem. Eng.* 2025, 13, 16303–16312



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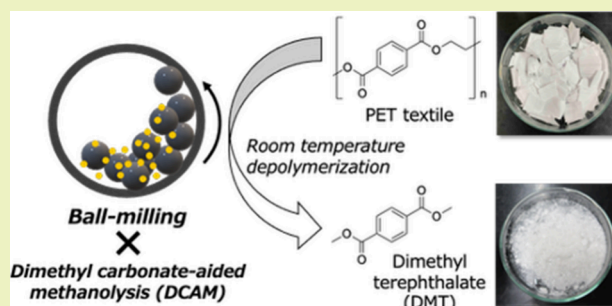
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**ABSTRACT:** Although polyethylene terephthalate (PET) bottles are conventionally reused using a material recycling process, the high content of impurities, such as dyes, pigments, and/or other fibers in the PET fibers, complicates the process. Chemical recycling is a promising method for PET fibers, but the depolymerization of PET fibers under mild conditions is still challenging due to their high crystallinity from the yarn-making process. Herein, we developed a rapid and efficient depolymerization process for PET fibers at room temperature by combining dimethyl carbonate (DMC)-aided methanolysis (DCAM) with a ball-milling (BM) approach. Compared with DCAM of PET fibers in a flask, the BM approach reduced not only the reaction time but also the amount of DMC required, which are key advantages toward practical application. To clarify the acceleration effect of BM on the depolymerization, PET fiber residues after the reaction were analyzed by gel permeation chromatography, differential scanning calorimetry, X-ray scattering, and solid-state NMR. The findings suggested that BM promotes the phase transformation of PET from its crystalline phase to its amorphous phase. In addition, BM induces efficient contact of the surface of the polymer with the catalyst/reagent, even in heterogeneous reaction media.

**KEYWORDS:** depolymerization, polyester, chemical recycling, ball-milling, methanolysis, catalysis



## INTRODUCTION

Social concerns regarding environmental/marine pollution caused by waste plastics have stimulated the quest to establish efficient plastic recovery and recycling methods.<sup>1–4</sup> Polyester fibers are the most widely used synthetic fibers, produced by spinning poly(ethylene terephthalate) (PET) resin. The annual worldwide production of polyester fibers is approximately 60 million tons per year, more than twice that of PET bottles.<sup>5</sup> A recent survey revealed that 87% of discarded polyester fabric products are incinerated or accumulated in landfills, with an estimated amount of ~42 million tons per year.<sup>6</sup> The main reason for the slow progress in recycling of fabric products is that they are blended with dyes and other fibers, making the material recycling method that is widely utilized for PET bottles practically inapplicable.<sup>7–13</sup> For this reason, recent studies on the issue have focused on the development of chemical recycling methods that can be applied to blended products.

Conventional chemical recycling methods for PET require the depolymerization process under high-temperature, high-pressure reaction conditions,<sup>14–18</sup> leading to high energy consumption as well as high CO<sub>2</sub> emission. Therefore, low-temperature depolymerization methods have recently attracted attention in the field of catalytic and biological processes. An

enzymatic method has emerged quite recently,<sup>16,19,20</sup> in which the depolymerization of PET proceeds at moderate temperature (~50 °C), giving terephthalic acid with high purity.<sup>21</sup> A drawback of this process is the relatively slow reaction rate, particularly for the crystalline component of PET. Therefore, increasing the amorphous component by pretreatment is necessary.<sup>22–24</sup> Alternatively, chemical approaches for low-temperature depolymerization of PET have also attracted recent attention.<sup>11,25–29</sup> In this context, we recently reported a method for the rapid depolymerization of PET with a high crystalline component (PET fiber) at a low temperature (~50 °C) by dimethyl carbonate (DMC)-aided methanolysis (DCAM).<sup>30</sup> This method allowed us to isolate dimethyl terephthalate (DMT) by trapping ethylene glycol with DMC to shift the transesterification equilibrium to a product system, but a higher reaction rate and a reduced amount of DMC are highly desired to realize a more cost-effective process.

**Received:** May 6, 2025

**Revised:** August 15, 2025

**Accepted:** August 21, 2025

**Published:** September 26, 2025



Table 1. Screening of Reaction Conditions for Depolymerization of Polyester Textile (100%, White)<sup>a</sup>

|                 | catalyst           | DMC (mL)       | MeOH (mL) | BM conditions                                     | DMT (%) | EG (%)       | EC (%) | DDH (%)      |
|-----------------|--------------------|----------------|-----------|---------------------------------------------------|---------|--------------|--------|--------------|
| 1               | NaOMe              | 0.2            | 0.02      | 10 mL jar (SUS), 10 $\phi$ ball x 1               | 83      | <1           | 65     | 8            |
| 2               | KOMe               | 0.2            | 0.02      | 10 mL jar (SUS), 10 $\phi$ ball x 1               | 80      | <1           | 61     | 8            |
| 3               | LiOMe              | 0.2            | 0.02      | 10 mL jar (SUS), 10 $\phi$ ball x 1               | 5       | 1            | 3      | <1           |
| 4               | TBD                | 0.2            | 0.02      | 10 mL jar (SUS), 10 $\phi$ ball x 1               | 24      | 0            | 20     | 3            |
| 5               | None               | 0.2            | 0.02      | 10 mL jar (SUS), 10 $\phi$ ball x 1               | 0       | 0            | 0      | 0            |
| 6               | NaOMe <sup>b</sup> | 0.2            | 0.02      | 10 mL jar (SUS), 10 $\phi$ ball x 1               | 92      | 4            | 63     | 8            |
| 7               | NaOMe              | 0              | 0.2       | 10 mL jar (SUS), 10 $\phi$ ball x 1               | 14      | <1           | 0      | 0            |
| 8               | NaOMe              | 0 <sup>c</sup> | 0.02      | 10 mL jar (SUS), 10 $\phi$ ball x 1               | 22      | <1           | 0      | <1           |
| 9               | NaOMe              | 0.2            | 0.02      | 10 mL jar (SUS), 5 $\phi$ ball x 10               | 85      | <1           | 65     | 9            |
| 10              | NaOMe              | 0.2            | 0.02      | 10 mL jar (ZrO <sub>2</sub> ), 10 $\phi$ ball x 1 | 73      | <1           | 54     | 8            |
| 11 <sup>d</sup> | NaOMe              | 0.2            | 0.02      | 10 mL jar (SUS), 10 $\phi$ ball x 1               | 70      | <1           | 53     | 7            |
| 12 <sup>e</sup> | NaOMe              | 0.2            | 0.02      | 2 mL Vial, 50 °C                                  | 23      | <sup>f</sup> | 32     | <sup>f</sup> |

<sup>a</sup>Conditions: PET textile (100 mg), catalyst (6 mol % based on PET alternating unit), DMC (0.2 mL, 4.6 eq based on PET alternating unit), MeOH (0.02 mL, 0.95 eq based on PET alternating unit), RT, 25 Hz, 10 min. Yields were determined by gas chromatography using biphenyl as a standard. Averages of >2 runs were shown. <sup>b</sup>NaOMe (20 mol %) was used. <sup>c</sup>Toluene (0.2 mL) was added. <sup>d</sup>PET flake (100 mg) was used instead of a PET textile. <sup>e</sup>Reaction time was 20 min. <sup>f</sup>Not determined.

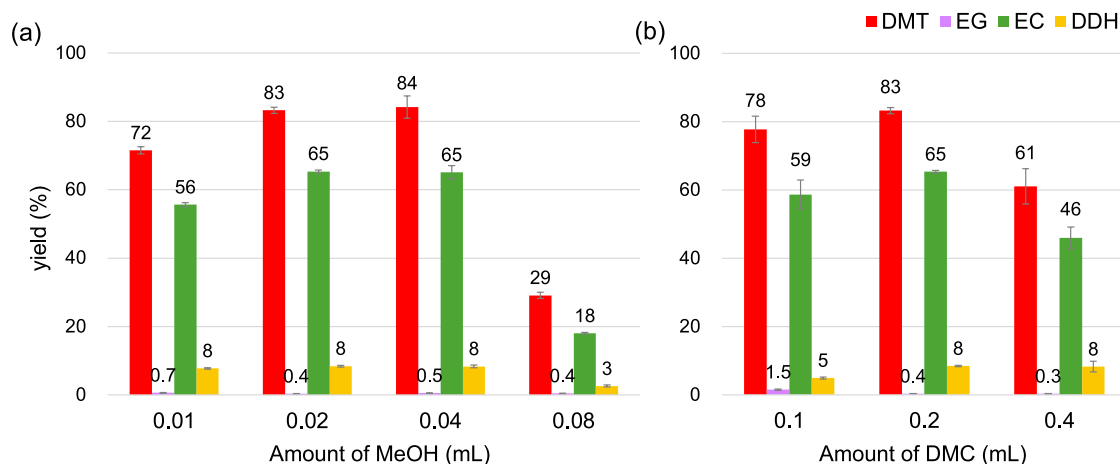
To this end, we focused on mechanochemical reactions using ball-milling (BM). It was recently demonstrated that various catalytic transformations of insoluble solids are enabled by utilizing BM instead of conventional reactions with a flask and solvents. The BM method promotes chemical reactions at the solid–solid interface due to the grinding force between the balls and the wall of the jar<sup>31–33</sup> and is promising for the depolymerization of polymeric materials under heterogeneous conditions.<sup>34–39</sup> Recent studies reported the depolymerization of PET by utilizing BM, but harsh conditions were required, such as hydrolysis reactions using an equimolar amount of base.<sup>40–43</sup> Sievers et al. reported that mechanochemical depolymerization of PET with 2 equiv of NaOH proceeded efficiently in 20 min, whereas an equimolar salt waste was produced after the workup process.<sup>42,43</sup> Auclair et al. studied the enzymatic depolymerization of crystalline PET combined with BM.<sup>44</sup> Depolymerization was promoted by BM pretreatment before the reaction, but a repeated BM/depolymerization process was required to achieve high monomer yield. More recently, Kim et al. reported mechanochemical depolymerization of powdered PET by methanolysis without any catalysts.<sup>45</sup> Although the reaction temperature inside was not monitored and discussed for PET depolymerization, it should be difficult to exclude the unexpected thermal effect on the catalyst-free depolymerization by long planetary milling time (650 rpm, 6 h).<sup>46</sup> Takagaki et al. also reported CaO-catalyzed methanolysis of powdered PET using planetary milling for a long time (600 rpm, 6 h).<sup>47</sup> Overall, it is still challenging to depolymerize the crystalline PET fiber in a short reaction time under mild and environmentally benign conditions.

In this study, by combining BM with DCAM, efficient depolymerization of PET fibers was achieved in a short period of time (15 min) using a reduced amount of DMC compared with that of the reaction in a flask. Analyses of the PET fibers

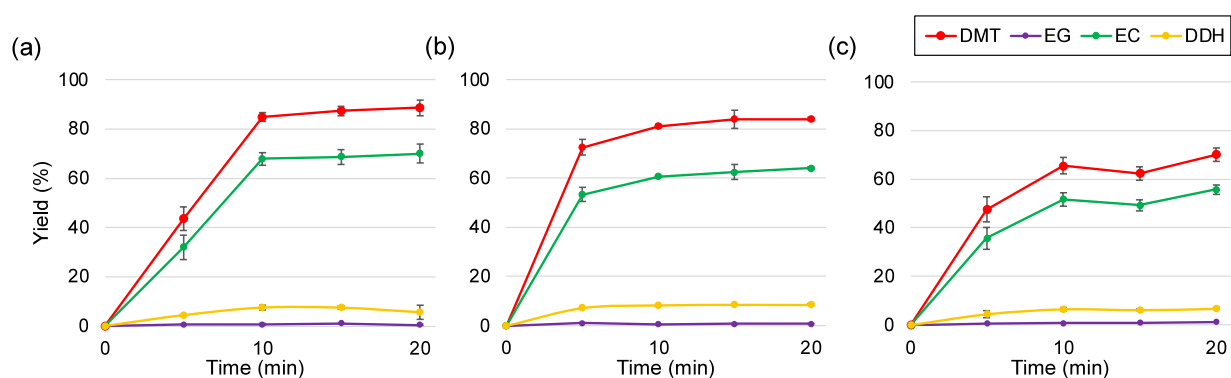
after the reaction using several methods to trace the higher-order structural changes of PET elaborated the acceleration effect of BM on the depolymerization of PET.

## RESULTS AND DISCUSSION

We began by using a PET textile obtained from a white lab coat (PET 100%) as a model starting material. The PET textile (100 mg) cut into ca. 10 mm squares was charged into a stainless-steel (SUS) jar (10 mL) with dimethyl carbonate (DMC; 0.2 mL), MeOH (0.02 mL), and base as a catalyst (6 mol % based on PET repeating unit: 192.17 g/mol). The reaction was carried out for 10 min at room temperature using a SUS ball (10  $\phi$ ) as the standard condition. A paste-like solid formed after the reaction was extracted with DMC, filtered, and the soluble components were quantified by gas chromatography (GC). Catalyst screening was performed first: dimethyl terephthalate (DMT) and ethylene carbonate (EC) were obtained in 83 and 63% yield, respectively, when NaOMe was used as a catalyst (entry 1). These yields were slightly decreased when KOMe was used instead of NaOMe (DMT: 80%, EC: 60%, entry 2). Dimethyl 2,5-dioxahexandionate (DDH) was also produced in 8% yield as a byproduct derived from ethylene glycol (EG) and DMC (entries 1, 2). The use of LiOMe or 1,5,7-triazabicyclo[4.4.0]dec-5-ene (TBD) as a catalyst decreased the yields (entries 3 and 4). The reaction did not proceed without a catalyst (entry 5). Higher loading of NaOMe (20 mol %) afforded higher DMT yield (92%) but lower EC (63%) compared to the run with NaOMe (6 mol %, entry 1). The yield was low when only 0.2 mL of MeOH was added without DMC or when toluene was used instead of DMC (entries 7, 8). Increasing the number of balls with a smaller size (5  $\phi$ ) improved the yield of DMC (entry 9). The use of a jar and a ball made of ZrO<sub>2</sub> material slightly reduced the yield (entry 10). PET flakes obtained from



**Figure 1.** Effect of DMC/MeOH ratio toward products yields. Screening of the (a) MeOH amount with 0.2 mL of DMC and (b) DMC amount with 0.02 mL of MeOH. Other reaction conditions were identical to those of Table 1, entry 1.



**Figure 2.** Time-course plot of products yields. The loading of NaOMe catalyst was 6 (a), 10 (b), and 3 mol % (c), respectively. Other reaction conditions were identical to those of Table 1, entry 1.

a PET bottle were also depolymerized to afford DMT in a 70% yield (entry 11). Slightly lower yield of DMT is probably due to a lower surface area of flake form than that of fiber form. To verify the heating effect of the BM, the internal temperature of the jar was measured with an infrared thermometer immediately after the reaction and determined to be 40 °C. We thus conducted the reaction at 50 °C under the same reaction conditions using a glass vial, resulting in 23% of DMT after 20 min (entry 11), which rules out the possibility that the reaction acceleration effect is simply due to a heating effect. Overall, we demonstrated that BM enabled a faster reaction rate and reduced the amount of DMC needed by 40% compared with the previously reported flask-type reaction (0.5 mL).<sup>29</sup>

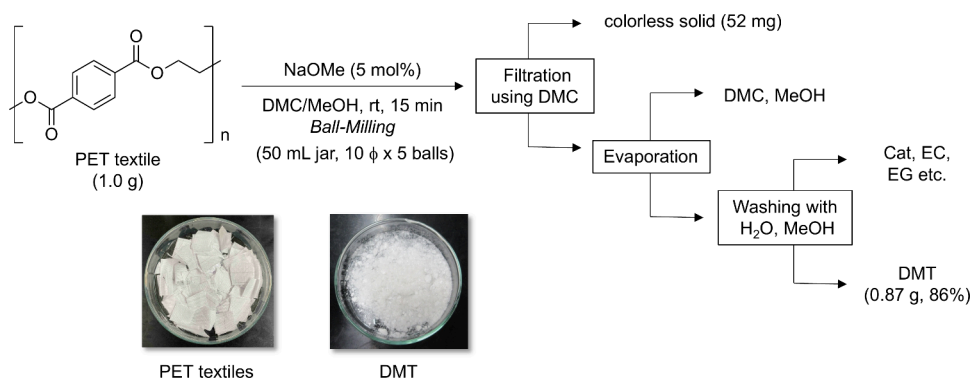
Next, the DMC/MeOH ratio for the reaction was optimized. The volume of MeOH was changed to a fixed volume of DMC at 0.2 mL (2 mL/g-PET). The highest yields of DMT (83%–84%) were established using 0.02–0.04 mL MeOH (Figure 1a). When the amount of MeOH was fixed at 0.02 mL (0.2 mL/g-PET) and the amount of DMC was changed, the maximum yield was obtained at 0.2 mL (Figure 1b). At any ratio, the yield of EG was less than 1%; the ratio of DDH tended to be improved at higher ratios of DMC.

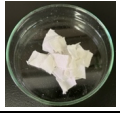
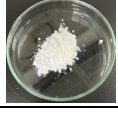
Under optimal reaction conditions, the product yields were plotted over time (Figure 2a): DMT and EC yields were almost saturated at >80% after 10 min reaction time, and after 20 min, yields reached 85% (DMT) and 65% (EC). EG yields

were low at all stages of the reaction, indicating that the generated EG was rapidly trapped by DMC. Compared with the reaction in a glass flask, the EC yield was lower by ~20% in the present reaction, and ~10% of DDH was produced instead. A time-course study was also conducted using higher (10 mol %) and lower (3 mol %) loading of NaOMe catalyst. DMT yield reached over 70% after only 5 min when 10 mol % NaOMe was loaded (Figure 2b), whereas a lower loading resulted in an insufficient DMT yield even after 20 min (Figure 2c). The composition of products derived from the EG moiety (EG, EC, and DDH) was mostly unchanged when the catalyst loading varied.

To clarify the reason for the reduced EC yield observed in this study, we conducted control experiments using EC as the starting material (Table S1). After BM treatment with NaOMe (0.1 equiv) in DMC/MeOH for 10 min, 40% of EC was consumed, and EG (5.5%), DDH (9.3%) were produced. Ring-opened products, such as 2-hydroxyethyl methyl carbonate and oligomers derived from EC, were also observed by NMR and GPC analyses. A similar result was obtained when the reaction was performed in a glass vial, indicating that these side reactions can occur regardless of the reaction setup. Indeed, our previous study on DCAM using a glass flask showed that an extended reaction time led to a decrease in EC yield.<sup>30</sup> Therefore, we assume that the BM reaction not only accelerated the depolymerization rate of PET but also

## Scheme 1. One Gram Scale Depolymerization of PET Textile Using BM Reaction

Table 2. Depolymerization of Blended Textiles with BM Reaction<sup>a</sup>

| Entry | PET textiles (wt%)                  |                                                                                     | Catalyst loading (mol% based on PET) | Recovered (wt%) | DMT Yield (g) <sup>b</sup> |                                                                                       |
|-------|-------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------|-----------------|----------------------------|---------------------------------------------------------------------------------------|
| 1     | PET/Cotton (65/35)                  |    | 15                                   | 33              | 0.53 g (74%)               |    |
| 2     | PET/Rayon/PU <sup>c</sup> (54/39/7) |    | 19                                   | 39              | 0.38 g (69%)               |    |
| 3     | Colored PET textile (100)           |  | 10                                   | <1              | 0.83 g (82%)               |  |

<sup>a</sup>Conditions: PET textile (1 g), NaOMe, DMC (1.5 mL), MeOH (0.2 mL), RT, 25 Hz, 15 min. <sup>b</sup>Isolated yields (g) after distillation are shown. The values in parentheses indicate molar yield (%) based on PET in each starting material. <sup>c</sup>Elastic polyurethane.

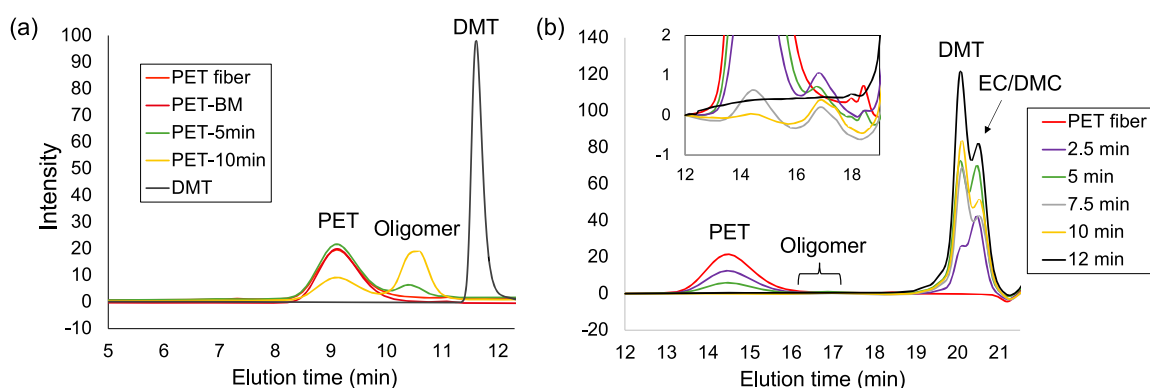
promoted side reactions involving EC, resulting in a lower EC yield even after a short reaction time.

A gram scale experiment, along with isolation of the products, was investigated using a 50 mL stainless-steel jar and five 10 $\phi$  balls. After the reaction for 15 min, the soluble components were extracted with CH<sub>2</sub>Cl<sub>2</sub> and concentrated under reduced pressure. The residue was washed with water and MeOH and dried to give 0.87 g of DMT (Scheme 1, 86% isolated yield). The obtained DMT was found to be pure based on <sup>1</sup>H NMR and GC analysis (Figures S1 and S2). The <sup>1</sup>H NMR spectrum of the insoluble solid (52 mg) obtained by the first filtration displayed predominantly disodium terephthalate and MeOH (Figure S3). The compound should be formed by the hydrolysis of DMT with NaOH, which is produced by NaOMe and residual water. By concentrating the water washings, a mixture of EC and DDH was obtained at a molar ratio of 1:0.11. To simplify the postreaction purification method, we also investigated a direct distillation of the paste-like solid product after the reaction. First, EC and DDH were separated at 100 °C under reduced pressure, and then, DMT was sublimed by increasing the temperature to 150 °C under reduced pressure. This method also yielded 0.90 g (89%) of DMT as colorless crystals. The overall process proved to be practical, as additional DMC and water were not required for the purification process. Considering that the reaction using enzymes takes longer than 2 days for the depolymerization of

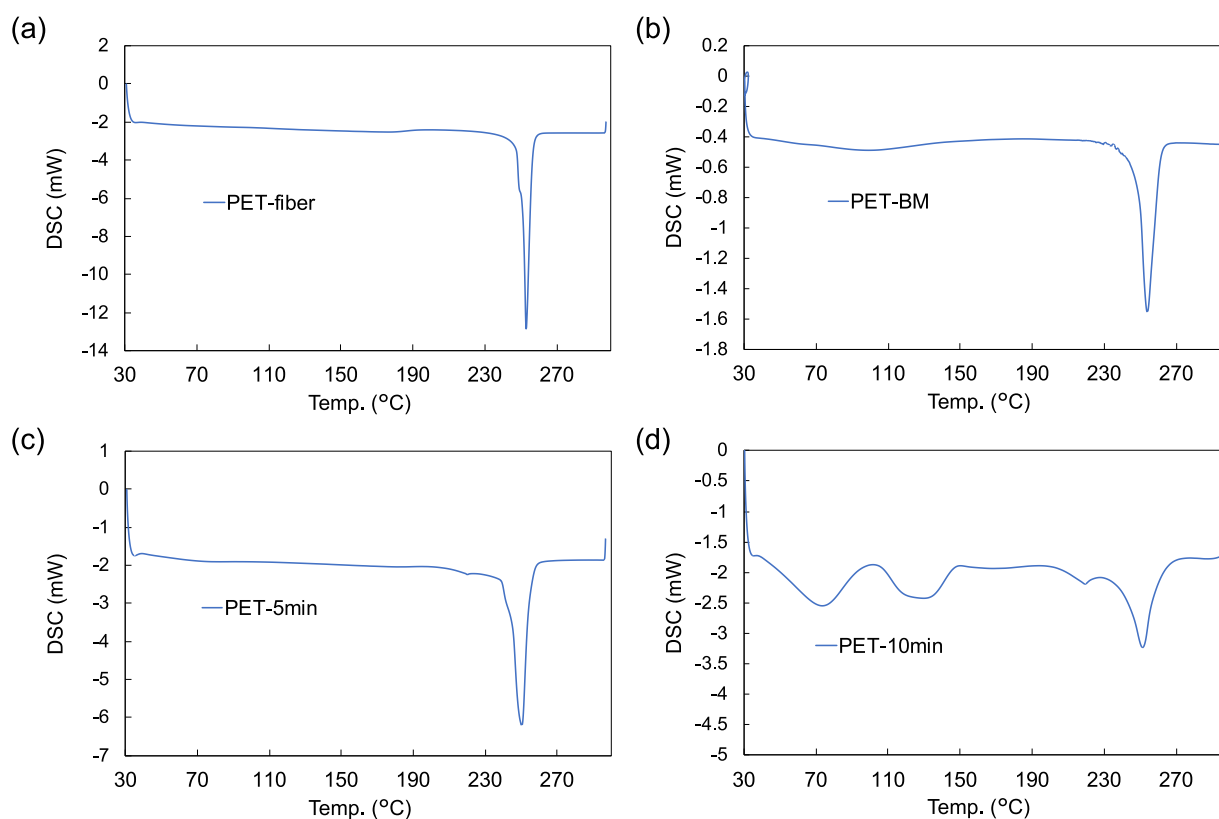
PET fiber,<sup>22</sup> and the hydrolysis of PET using BM requires more than 1 h;<sup>40</sup> the current reaction provides a competitive low-temperature method, enabling the depolymerization of PET fibers at a faster rate to isolate pure DMT in high yield.

To discuss the scope of the starting materials, including PET, we chose the blended textiles. The results for each textile are summarized in Table 2. Although a higher catalyst loading (15 mol %) was needed compared to an experiment with genuine PET textile (100%), crystalline DMT was obtained from PET/Cotton (65/35) blend in 74% yield after subliming crude DMT at 120 °C under reduced pressure (Table 2, entry 1). Similarly, crystalline DMT was obtained in 69% yield from PET/Rayon/PU (54/39/7) blend using 19 mol % catalyst (Table 2, entry 2). However, the depolymerization did not proceed completely even with the higher catalyst loading condition because both FT-IR (Figure S4) and solid-state NMR spectra (Figure S5) of the unreacted solids obtained from blended textiles displayed the contamination of unreacted PET (~10% for entry 1, ~9% for entry 2). Depolymerization of colored PET textiles also afforded a slightly lower yield of crystalline DMT (82%) along with unreacted PET (<1%) using 10 mol % catalyst. Those results implied that the activity of the NaOMe catalyst is sensitive to impurities under BM conditions. In fact, DCAM of blended PET textiles (i.e., PET/Cotton, PET/PU, and PET/dye) using a glass flask with DMC as a solvent at 50 °C proceeded more efficiently.<sup>30</sup> We consider





**Figure 3.** GPC profiles of PET fiber (red), PET-BM (green), PET-5 min (orange), PET-10 min (yellow), and DMT (black) (a). Time-course GPC profiles of PET fiber (red) and crude product after 2.5 (purple), 5 (green), 7.5 (gray), 10 (yellow), and 12 min (black) BM reaction (b). Note that profiles shown in (a) and (b) were measured under nonidentical GPC conditions: eluent used in (a) was HFIP, and that in (b) was HFIP with CF<sub>3</sub>COONa (10 mM), resulting in different elution time of each component.

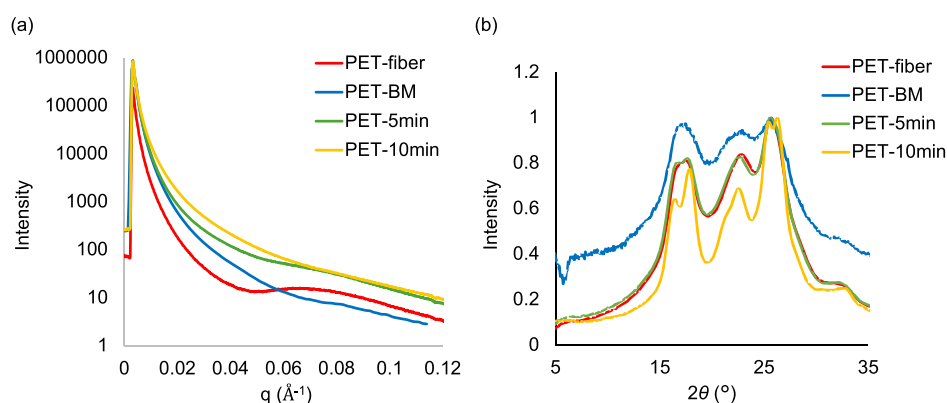


**Figure 4.** DSC curve (first heating) of PET fibers (a), PET-BM (b), PET-5 min (c), and PET-10 min (d).

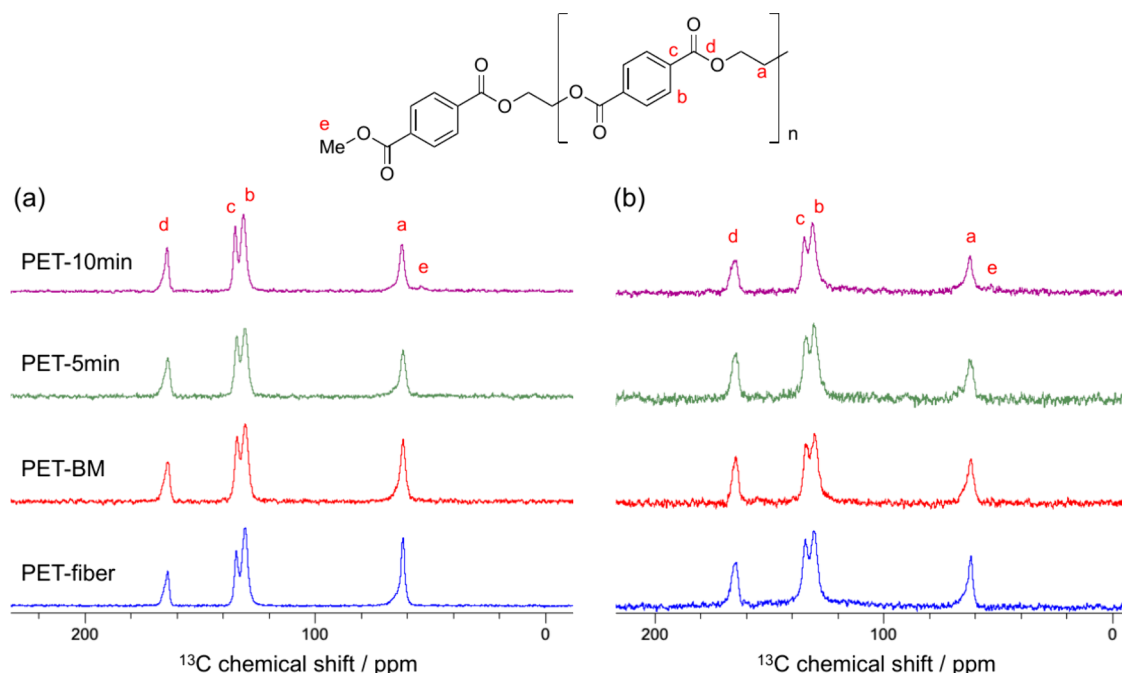
that the reason for the decrease in the reaction efficiency in BM should be the deactivation of the catalyst. Since a BM treatment with a small amount of DMC might increase the contact of the catalyst with impurities as well as PET, a faster deactivation of the catalyst, rather than that in the liquid phase reaction with a glass flask, should be induced. Eventually, we demonstrated that a substantial amount of analytically pure DMT (Figures S6–11) from blended PET textiles was produced in a short reaction time and isolated with a simple purification process.

Our next concern was the acceleration mechanism by BM. Four samples were prepared under different conditions to monitor the reaction progress for the mechanistic study: 1) before the reaction, 2) after 5 min depolymerization (PET-5

min), 3) after 10 min depolymerization (PET-10 min), and 4) PET textile ball-milled in DMC/MeOH for 15 min without catalyst (PET-BM) (from 1.0 g of PET textile, 0.44 g of PET residue in the 5 min reaction, 0.13 g of PET residue in the 10 min reaction). PET-5 min and PET-10 min were purified by rinsing with DMC and water after the BM reaction in order to remove monomers and catalysts possibly attached to residual PET. First, the molecular weight was measured by gel permeation chromatography (GPC) using 1,1,1,3,3,3-hexafluoro-2-propanol (HFIP) as eluent (Figure 3a). Before the reaction, PET showed a weight-average molecular weight ( $M_w$ ) of  $3.1 \times 10^4$  g/mol, which remained almost unchanged after BM ( $3.0 \times 10^4$  g/mol). This result indicates that BM alone does not lead to mechanical force cleavage of the polymer



**Figure 5.** Small-angle X-ray scattering (SAXS) (a) and wide-angle X-ray scattering (WAXS) (b) measurements of PET fiber, PET-BM, PET-5 min, and PET-10 min.

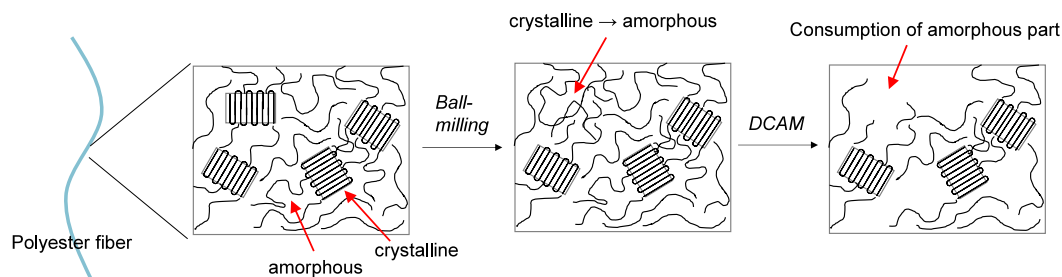


**Figure 6.** Solid-state  $^{13}\text{C}$  MAS NMR of PET materials using cross-polarization (CP) conditions (a) and direct polarization with high-power  $^1\text{H}$  decoupling conditions (b).

chain. Smaller molecular weight peaks were detected in the PET-5 min and PET-10 min, along with a high molecular weight peak similar to that of the raw material ( $M_w$ :  $2.0 \times 10^3$  g/mol [PET-5 min],  $1.8 \times 10^3$  g/mol [PET-10 min]). The area ratio of the low molecular weight peak increased after reacting for 10 min. This suggests the presence of oligomers corresponding to intermediates in this reaction. Because the GPC measurements of PET residue samples (PET-5 min and PET-10 min) excluded the observation of soluble oligomeric products as well as monomers (DMT, EC, etc.) due to the purification prior to the measurement, we additionally conducted the time-course GPC measurement of crude products to gain insight into the reaction mechanism. Crude products after BM reaction for 2.5, 5, 7.5, 10, and 12 min were dissolved in HFIP and employed for GPC measurements (Figure 3b). The peak intensity of PET (elution time: 14.5 min) gradually decreased, and that of DMT (elution time: 20 min) increased instead, indicating most of the consumed PET was very quickly converted into DMT monomer without the formal observation of oligomers. Only a closer look showed the

presence of small peaks assignable to oligomers (elution time: 16.5–17.5 min) in the profiles of crude product after 2.5–10 min BM reaction (Figure 3b inset), which is consistent with the measurement of purified PET residues (Figure 3a). Those peaks eventually disappeared in the GPC profiles of the crude product after 12 min, suggesting that longer BM treatment time allowed depolymerization of even such oligomers.

Next, differential scanning calorimetry (DSC) measurements were performed, and melting peaks were detected at  $252^\circ\text{C}$  for the raw material and at  $254^\circ\text{C}$  for PET-BM (Figure 4). The crystallinity calculated from the melting peak area was 40% for the raw material and 35% for PET-BM, indicating a slight decrease in crystallinity. This phenomenon is in line with a reported study, where the apparent crystallinity of a high-crystallinity PET decreased to  $\sim 36\%$  after milling.<sup>48</sup> The crystallinity of PET-5 min was determined to be 53%, indicating an increase. On the other hand, from PET-10 min, several peaks were observed in addition to the endothermic peak around  $250^\circ\text{C}$  derived from PET, making it difficult to calculate the crystallinity derived from pure PET.

**Scheme 2. Plausible Mechanism of DCAM Reaction Combined with the BM Approach in a View of a Macromolecular Scale**

Small-angle X-ray scattering (SAXS) measurements were performed on four different samples (Figure 5a). In PET-BM, a lamella-derived peak disappeared, but in PET-5 min, the peak was observed at the same position as that of PET fibers, suggesting that the higher-order structure of the crystalline layer remains unchanged, even though PET is depolymerized in the early stage of the reaction. The peak disappeared in the spectrum of PET-10 min. Next, wide-angle X-ray scattering (WAXS) measurements were performed (Figure 5b), and the crystallinity was calculated from the PET-derived peaks by deconvolution analysis as follows: PET-fiber 35%, PET-BM 25%, PET-5 min 29%, and PET-10 min 43%. These results indicate that the crystalline component decreases with BM, but the amorphous component disappears due to depolymerization as the reaction proceeds, resulting in a relative increase in the crystalline component.

Finally, local structural changes were examined by solid-state NMR measurements. First, structural analysis of each sample was performed by  $^{13}\text{C}$  cross-polarization magic-angle spinning (CPMAS) measurement (Figure 6a). Signals at 64 ppm (a) from the EG moiety, 130 (b) and 136 ppm (c) from the aromatic moiety, and 165 ppm (d) from the carbonyl moiety were observed in all samples, indicating that BM causes no local structural changes. The spectra of PET-5 min were almost identical with those of the raw materials. On the other hand, in addition to the PET-derived signal from PET-10 min, a new signal was observed at 54 ppm (e). This signal likely derives from the methyl ester and can be assigned to the end-group of the oligomer generated by the depolymerization. Next, the signal was quantitatively analyzed by  $^{13}\text{C}$  direct polarization with high-power  $^1\text{H}$  decoupling conditions (Figure 6b).<sup>49</sup> Based on the integral ratio of the observed signal of the EG moieties (e), the crystallinity was calculated to be PET-fiber 51%, PET-BM 31%, PET-5 min 48%, and PET-10 min 72%, respectively (Figure S12).

We also investigated the role of the solvent (DMC/MeOH) during the BM treatment. To elucidate the effect of DMC/MeOH, a control sample (PET-BM') was additionally prepared by subjecting the PET fiber to a BM treatment in the absence of both catalyst and DMC/MeOH. The sample also exhibited a decrease in crystallinity, as determined by DSC (39% (PET-BM') vs 40% (PET-fiber)), WAXS (11% (PET-BM') vs 35% (PET-fiber)), and solid-state  $^{13}\text{C}$  NMR measurement (47% (PET-BM') vs 51% (PET-fiber)). This result suggests that the use of DMC/MeOH does not significantly affect the phase transition during the BM treatment. A related study regarding bead milling of PET reported that the addition of toluene as a solvent led to a decrease in  $M_w$  of PET, thereby enabling methanolysis of PET at a low temperature (90 °C).<sup>41</sup> In contrast, in our study, the  $M_w$  of PET remained unchanged after BM with DMC/MeOH

(Figure 3a). Moreover, when toluene/MeOH was used as a solvent instead of DMC/MeOH, the yield of DMT was lower (22%; Table 1, entry 7). These findings indicated that the primary role of DMC in this system is to act as an EG trapping agent during catalytic methanolysis. The role of DMC in our system also differed from its use in the depolymerization of polycarbonate (PC) via planetary ball-milling,<sup>45</sup> where DMC served as a swelling solvent to facilitate grinding PC pellets into powder before depolymerization with MeOH.

Based on the results of the above structural analysis of PET, the BM reaction promotes the transition from the crystalline phase to the amorphous phase (Scheme 2). The apparent increase in crystallinity of the residual PET textiles during the reaction is likely due to the decrease in amorphous PET, which is preferentially depolymerized by the DCAM reaction at room temperature. In the late stage of the reaction, oligomers with low reactivity due to high crystallinity remain unreacted, resulting in a slower depolymerization reaction rate. The degree of crystallinity during PET depolymerization with BM treatment was also discussed with DSC analysis by Auclair et al. for an enzymatic reaction.<sup>44</sup> They reported that PET crystallinity did not vary significantly after depolymerization and concluded that both amorphous and crystalline regions were depolymerized to the same extent. Because our reaction system is not an enzymatic but a chemically catalytic one, and the protocol for the depolymerization is not identical (only BM in our system to produce DMT vs repetitive BM and aging at 50 °C to obtain TPA in Auclair's system), those systems were not simply comparable to each other. Noteworthy is that the PET conversion of our system (56% for PET-5 min and 87% for PET-10 min) is much higher than that of Auclair's work (TPA yield: 16%), allowing us to observe the transition of PET crystallinity more clearly by multiple analytical methods.<sup>44</sup>

Regarding the effect of BM, it should be noted that BM may increase the contact between the polymer surface and the catalyst and reactant, thereby improving the reaction efficiency. In fact, we observed a slight increase in the yield when a large number of balls were used (Table 1, entry 9).

Finally, we quantitatively evaluated the BM reaction and the previously reported flask-type reaction<sup>30</sup> in terms of energy consumption, as part of sustainability metrics.<sup>53</sup> For comparison, we separately assessed the energy consumption for PET depolymerization and solvent distillation in both processes (Table S2). Using an electrical power meter, the energy consumption (Wh) for PET depolymerization via the BM reaction (25 kHz, for 15 min, RT, 86% yield) was estimated as 64.5 Wh per gram of isolated DMT (Wh/g(DMT)), whereas the flask-type reaction (1000 rpm magnetic stirring, 2 h, 50 °C (oil bath), 94% yield) required 95 Wh/g(DMT). Additionally, based on the latent heat of DMC (1.2 kJ/g) and MeOH (0.50

kJ/g), the estimated energy required for solvent (DMC/MeOH) distillation was 1.4 kJ/g(DMT) for the BM reaction and 9.0 kJ/g(DMT) for the flask-type reaction. These results demonstrated that the BM reaction presented in this study offers a clear advantage over previous methods in terms of energy efficiency.

## CONCLUSIONS

In this study, we found that by combining the DCAM method with BM, PET fibers can be depolymerized in a short period of time (15 min.). In addition, the amount of DMC used could be reduced to 40% of that used in a previous study with a glass flask, indicating substantial progress from a practical standpoint. Direct vacuum distillation of the mixture after the reaction afforded crystalline DMT as a residue in an 89% yield. Blended PET textiles were also examined in this approach, affording crystalline DMT in slightly lower yield (65–74%), presumably due to a deactivation of the catalyst by impurities. To verify the reaction acceleration effect of the BM, *ex situ* analysis of PET fibers during the reaction was conducted by applying GPC, DSC, SAXS, WAXS, and solid-state NMR. The BM continuously promoted the transition from the crystalline phase to the amorphous phase, efficiently accelerating the depolymerization in a short period of time at room temperature. These findings pave the way for the low-energy chemical recycling of PET fibers. The detailed study on postconsumer textiles, including the effect of impurities in the textiles, is underway in our laboratory.

## EXPERIMENTAL SECTION

**1. General.** Polyester fibers used in this study were obtained from commercially available textiles (white lab coat for analytical experiment [PET 100%], white lab coat for chemical experiment [PET/cotton 65/35], functional undershirt [PET/Rayon/PU 54/39/7], and colored apron [PET 100%]). The textiles were cut with scissors into ca. 1 cm × 1 cm size and used for experiments as is. Chemicals were purchased from chemical suppliers. DMC, MeOH, and toluene were dried over activated MS4A before use. Retsche MM400 was used for the ball-milling reaction. Solution NMR was measured with a Bruker Avance III spectrometer (600 MHz for  $^1\text{H}$  nuclei) using a cryo probe. All spectra were recorded at  $25 \pm 1^\circ\text{C}$ . Chemical shifts ( $\delta$ ) are in parts per million relative to tetramethylsilane at 0 ppm for  $^1\text{H}$ . Gas chromatographic (GC) analyses were performed on a Shimadzu GC-2014 instrument using a DB-WAXETR column (0.25 mm × 30 m, Agilent Technologies). All samples were analyzed and quantified by using biphenyl as an internal standard. FT-IR spectroscopy was measured with Bruker ALPHA II using an attenuated total reflection (ATR) apparatus. Solid-state NMR was measured with an Avance NEO spectrometer (400 MHz for  $^1\text{H}$  nuclei) using a 3.2 mm MAS probe. Standard ramped cross-polarization (CP) was used to transfer the polarization from the  $^1\text{H}$  nuclei to the nucleus of interest ( $^{13}\text{C}$ ). SPINAL-64  $^1\text{H}$  heteronuclear decoupling was applied during the acquisition.  $^{13}\text{C}$  chemical shifts were referenced to tetramethylsilane at 0 ppm by using adamantane as an external standard (38.52 ppm).

**2. Procedure for the Catalytic Reaction. General Procedure for Optimization of Conditions.** In the jar (volume: 10 mL, stainless-steel or  $\text{ZrO}_2$ ), polyester textile (100 mg, 0.52 mmol based on an alternating unit) and the measured amount of DMC were charged. The catalyst (6 mol %) dissolved in MeOH, and balls ( $10\phi \times 1$  or  $5\phi \times 10$ , stainless-steel or  $\text{ZrO}_2$ ) were added. The jar was tightly closed and placed in the ball-milling apparatus and milled at 25 Hz for 10–20 min. The resulting paste-like solid was dissolved with DMC (2 mL × 2) and filtered using a syringe equipped with a PTFE filter (pore size: 0.22  $\mu\text{m}$ ). To a combined solution was then added a measured amount of biphenyl (as an internal standard), and an aliquot was used

for GC-FID measurement. Yields of DMT, EG, EC, and DDH were determined by the standard calibration curve method.

**Gram Scale Reaction for Colorless PET Textiles.** In the jar (volume: 50 mL, stainless-steel), PET textile (1.0 g, 5.2 mmol based on an alternating unit) and the measured amount of DMC were charged. NaOMe (5 mol %) dissolved in MeOH, and balls ( $10\phi \times 5$ , stainless-steel) were added. The jar was tightly closed, placed in the ball-milling apparatus, and milled at 25 Hz for 15 min. The resulting paste-like solid was dissolved with  $\text{CH}_2\text{Cl}_2$  (25 mL) and filtered using filter paper. The solution was concentrated under reduced pressure, and the residue was washed with  $\text{H}_2\text{O}$  (50 mL). DMT was obtained as a crystalline solid (0.87 g, 86% yield). Alternatively, a paste-like solid was simply dried and sublimed using a Kugelrohr apparatus. The mixture of EC, DDH (0.48 g), and DMT (0.90 g) was isolated successively.

**Gram Scale Reaction for Blended PET Textiles.** In the jar (volume: 50 mL, stainless-steel), blended PET textile (1.0–1.1 g) and the measured amount of DMC were charged. NaOMe (10–19 mol %) dissolved in MeOH, and balls ( $10\phi \times 5$ , stainless-steel) were added. The jar was tightly closed and placed in the ball-milling apparatus, and milled at 25 Hz for 15 min. The resulting paste-like solid was dissolved with  $\text{CH}_2\text{Cl}_2$  (25–30 mL) and filtered using filter paper. The solution was concentrated under reduced pressure, and the residue was washed with  $\text{H}_2\text{O}$  (50 mL). Crude DMT was further sublimed using a Kugelrohr apparatus. DMT was isolated as crystalline solids (65–74% yield, listed in Table 2). The insoluble solids were washed with  $\text{H}_2\text{O}$  (20 mL) to remove catalyst and byproducts from DMT (mainly disodium terephthalate). After drying at  $50^\circ\text{C}$  under reduced pressure, those solids were analyzed by FT-IR and solid-state NMR (Figures S4 and S5).

**Gel Permeation Chromatography (GPC).** GPC measurement was performed at  $40^\circ\text{C}$  with a Jasco system equipped with a PU-4185 RHPLC semimicro pump, JASCO AS-4150 RHPLC Autosampler, RI-4035 semimicro refractive index detector, CO-4065/4060 column oven, Gastorr FG-32 degasser, and TSKgel SuperHM-H columns (Tosoh Co., Tokyo, Japan). 1,1,1,3,3,3-Hexafluoro-2-propanol (HFIP) or HFIP containing  $\text{CF}_3\text{COONa}$  (10 mM) was used as an eluent at 0.3 mL  $\text{min}^{-1}$ . Data were recorded and analyzed with ChromNAV GPC software, version 2. Molecular weights ( $M_n$ ,  $M_w$ ) were determined relative to poly(methyl methacrylate) standards. For sample preparation for GPC, about 1 mg of PET was dissolved in 1 mL of HFIP. The solution was filtered through a polytetrafluoroethylene membrane with a pore size of 0.20  $\mu\text{m}$ . Then 20  $\mu\text{L}$  of the sample solution was injected into the GPC system.

**Differential Scanning Calorimetry (DSC).** Differential scanning calorimetry was performed using a DSC-7000X (Hitachi High-Tech, Japan) in an atmosphere of nitrogen flowing at 50 mL/min. A heating rate of  $10^\circ\text{C min}^{-1}$  was applied for the temperature range from 30 to  $300^\circ\text{C}$ . The degree of crystallinity ( $X_c$ ) was calculated from the area of the DSC endotherm of the first heating scan using 140 J/g as the enthalpy of fusion of a 100% crystalline PET.<sup>50–52</sup> The crystallinity can be obtained from eq 1:

$$X_c = \frac{\Delta H_m}{\Delta H_f} \quad (1)$$

where  $\Delta H_m$  is the measured enthalpy of fusion of PET samples and  $\Delta H_f$  is the enthalpy of fusion of a 100% crystalline PET from the literature.

**SAXS and WAXS Measurement.** Two-dimensional (2D) WAXS and SAXS measurements were performed using Cu  $K\alpha$  radiation and a NANOPIX instrument (40 kV and 30 mA; Rigaku, Tokyo, Japan) equipped with a HyPix-6000C counting pixel detector (pixel size of  $100 \times 100 \mu\text{m}^2$ ) at room temperature. Camera distances were calculated using crystalline silicon (WAXS) or silver behenate (SAXS). The obtained 2D scattering patterns were converted into one-dimensional (1D) scattering intensity profiles, with radially averaged intensity calculated as a function of scattering vector  $q = (4\pi\sin(\theta/2))/\lambda$  on SAXS pattern and  $2\theta$  on WAXS pattern;  $\theta$  is the



scattering angle, and  $\lambda$  is the wavelength of the incident X-ray (0.154 nm in Cu K $\alpha$  radiation).

## ■ ASSOCIATED CONTENT

### SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acssuschemeng.5c04227>.

Details of control experiments using ethylene carbonate (EC) as a starting material (Table S1), NMR and GC spectra of isolated products from PET textiles (100%) (Figure S2), NMR analysis of insoluble part after depolymerization (Figure S3), analysis of products obtained from DCAM of blended textiles (Figures S4–S11), determination of crystallinity based on  $^{13}\text{C}$  solid-state NMR spectroscopy (Figure S12), and evaluation of the energy consumption (Table S2) (PDF)

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## Notes

The authors declare no competing financial interest.

## ■ ACKNOWLEDGMENTS

This work was supported by the Environment Research and Technology Development Fund (3RF-2302) of the Environmental Restoration and Conservation Agency, provided by the Ministry of the Environment of Japan, and the JACI Prize for Encouraging Young Researchers.

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